Chapter P7: Ideas about Science

Overview

In order to make sense of the scientific ideas that learners encounter in lessons and in everyday life outside of school, they need an understanding of how science explanations are developed, the kinds of evidence and reasoning behind them, their strengths and limitations, and how far we can rely on them.

Learners also need opportunities to consider the impacts of science and technology on society, and how we respond individually and collectively to new ideas, artefacts and processes that science makes possible.

It is intended that the *Ideas about Science* will help learners understand how scientific knowledge is obtained, how to respond to science stories and issues in the world outside the classroom, and the impacts of scientific knowledge on society.

Note that:

- although particular *Ideas about Science* have been linked to particular contexts throughout the specification as examples, the assessable learning outcomes in this chapter should be developed, and will be assessed, in any context from chapters P1–P6
- the assessable learning outcomes in this chapter will be assessed in all of the written examination papers
- terms associated with measurement and data analysis are used in accordance with their definitions in the Association of Science Education publication *The Language of Measurement* (2010).

Learning about How Science Works before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- understand that science explanations are based on evidence and that as new evidence is gathered, explanations may change
- have devised and carried out scientific enquiries, in which they have selected the most appropriate techniques and equipment, collected and analysed data and drawn conclusions.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

IaS1 What needs to be considered when investigating a phenomenon scientifically?

<table>
<tr><th>Teaching and learning narrative</th><th colspan="2">Assessable learning outcomes
Learners will be required to:</th><th>Linked learning opportunities</th></tr>
<tr><td rowspan="8">The aim of science is to develop good explanations for natural phenomena. There is no single ‘scientific method’ that leads to good explanations, but scientists do have characteristic ways of working. In particular, scientific explanations are based on a cycle of collecting and analysing data.

Usually, developing an explanation begins with proposing a hypothesis. A hypothesis is a tentative explanation for an observed phenomenon (“this happens because . . .”).

The hypothesis is used to make a prediction about how, in a particular experimental context, a change in a factor will affect the outcome. A prediction can be presented in a variety of ways, for example in words or as a sketch graph.

In order to test a prediction, and the hypothesis upon which it is based, it is necessary to plan an experimental strategy that enables data to be collected in a safe, accurate and repeatable way.</td><td>1.</td><td>in given contexts use scientific theories and tentative explanations to develop and justify hypotheses and predictions</td><td rowspan="8"></td></tr>
<tr><td>2.</td><td>suggest appropriate apparatus, materials and techniques, justifying the choice with reference to the precision, accuracy and validity of the data that will be collected</td></tr>
<tr><td>3.</td><td>recognise the importance of scientific quantities and understand how they are determined</td></tr>
<tr><td>4.</td><td>identify factors that need to be controlled, and the ways in which they could be controlled</td></tr>
<tr><td>5.</td><td>suggest an appropriate sample size and/or range of values to be measured and justify the suggestion
M2d</td></tr>
<tr><td>6.</td><td>plan experiments or devise procedures by constructing clear and logically sequenced strategies to:
– make observations
– produce or characterise a substance
– test hypotheses
– collect and check data
– explore phenomena</td></tr>
<tr><td>7.</td><td>identify hazards associated with the data collection and suggest ways of minimizing the risk</td></tr>
<tr><td>8.</td><td>use appropriate scientific vocabulary, terminology and definitions to communicate the rationale for an investigation and the methods used using diagrammatic, graphical, numerical and symbolic forms</td></tr>
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IaS2 What processes are needed to draw conclusions from data?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
The cycle of collecting, presenting and analysing data usually involves translating data from one form to another, mathematical processing, graphical display and analysis; only then can we begin to draw conclusions. A set of repeat measurements can be processed to calculate a range within which the true value probably lies and to give a best estimate of the value (mean). Displaying data graphically can help to show trends or patterns, and to assess the spread of repeated measurements. Mathematical comparisons between results and statistical methods can help with further analysis.	1. present observations and other data using appropriate formats 2. when processing data use SI units where appropriate (e.g. kg, g, mg; km, m, mm; kJ, J) and IUPAC chemical nomenclature unless inappropriate 3. when processing data use prefixes (e.g. tera, giga, mega, kilo, centi, milli, micro and nano) and powers of ten for orders of magnitude 4. be able to translate data from one form to another M2c, M4a 5. when processing data interconvert units 6. when processing data use an appropriate number of significant figures M2a 7. when displaying data graphically select an appropriate graphical form, use appropriate axes and scales, plot data points correctly, draw an appropriate line of best fit, and indicate uncertainty (e.g. range bars) M2c, M4a, M4c 8. when analysing data identify patterns/trends, use statistics (range and mean) and obtain values from a line on a graph (including gradient, interpolation and extrapolation), M2b, M2f, M2g, M4b, M4d, M4e, M4f	P6.2 (mechanical equivalent of heat) Describe and explain how careful experimental strategy can yield high quality data.

IaS2 What processes are needed to draw conclusions from data?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Data obtained must be evaluated critically before we can make conclusions based on the results. There could be many reasons why the quality (accuracy, precision, repeatability and reproducibility) of the data could be questioned, and a number of ways in which they could be improved. Data can never be relied on completely because observations may be incorrect and all measurements are subject to uncertainty (arising from the limitations of the measuring equipment and the person using it). A result that appears to be an outlier should be treated as data, unless there is a reason to reject it (e.g. measurement or recording error)	9. in a given context evaluate data in terms of accuracy, precision, repeatability and reproducibility, identify potential sources of random and systematic error, and discuss the decision to discard or retain an outlier 10. evaluate an experimental strategy, suggest improvements and explain why they would increase the quality (accuracy, precision, repeatability and reproducibility) of the data collected, and suggest further investigations	
Agreement between the collected data and the original prediction increases confidence in the tentative explanation (hypothesis) upon which the prediction is based, but does not prove that the explanation is correct. Disagreement between the data and the prediction indicates that one or other is wrong, and decreases our confidence in the explanation.	11. in a given context interpret observations and other data (presented in diagrammatic, graphical, symbolic or numerical form) to make inferences and to draw reasoned conclusions, using appropriate scientific vocabulary and terminology to communicate the scientific rationale for findings and conclusions 12. explain the extent to which data increase or decrease confidence in a prediction or hypothesis	

IaS3 How are scientific explanations developed?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Scientists often look for patterns in data as a means of identifying correlations that can suggest cause-effect links – for which an explanation might then be sought. The first step is to identify a correlation between a factor and an outcome. The factor may then be the cause, or one of the causes, of the outcome. In many situations, a factor may not always lead to the outcome, but increases the chance (or the risk) of it happening. In order to claim that the factor causes the outcome we need to identify a process or mechanism that might account for the observed correlation.	1. use ideas about correlation and cause to: – identify a correlation in data presented as text, in a table, or as a graph M2g – distinguish between a correlation and a cause-effect link – suggest factors that might increase the chance of a particular outcome in a given situation, but do not invariably lead to it – explain why individual cases do not provide convincing evidence for or against a correlation – identify the presence (or absence) of a plausible mechanism as reasonable grounds for accepting (or rejecting) a claim that a factor is a cause of an outcome	*Considering correlation and cause:* evidence for risks of X-rays (P1.2) evidence for human activities causing global warming (P1.3)
Scientific explanations and theories do not 'emerge' automatically from data, and are separate from the data. Proposing an explanation involves creative thinking. Collecting sufficient data from which to develop an explanation often relies on technological developments that enable new observations to be made. As more evidence becomes available, a hypothesis may be modified and may eventually become an accepted explanation or theory. A scientific theory is a general explanation that applies to a large number of situations or examples (perhaps to all possible ones), which has been tested and used successfully, and is widely accepted by scientists. A scientific explanation of a specific event or phenomenon is often based on applying a scientific theory to the situation in question.	2. describe and explain examples of scientific methods and theories that have developed over time and how theories have been modified when new evidence became available	*Developing scientific explanations:* Climate change (P1.3) **Big Bang model (P4.5)** Nuclear model of the atom (P5.1) The link between work, heat and temperature (P6.2)

IaS3 How are scientific explanations developed?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Findings reported by an individual scientist or group are carefully checked by the scientific community before being accepted as scientific knowledge. Scientists are usually sceptical about claims based on results that cannot be reproduced by anyone else, and about unexpected findings until they have been repeated (by themselves) or reproduced (by someone else). Two (or more) scientists may legitimately draw different conclusions about the same data. A scientist's personal background, experience or interests may influence his/her judgments. An accepted scientific explanation is rarely abandoned just because new data disagree with it. It usually survives until a better explanation is available.	3. describe in broad outline the 'peer review' process, in which new scientific claims are evaluated by other scientists	*Explanations that relied on technological development:* **Telescopes and the Big Bang model (P4.5)**
Models are used in science to help explain ideas and to test explanations. A model identifies features of a system and rules by which the features interact. It can be used to predict possible outcomes. Representational models use physical analogies or spatial representations to help visualise scientific explanations and mechanisms. Descriptive models are used to explain phenomena. Mathematical models use patterns in data of past events, along with known scientific relationships, to predict behaviour; often the calculations are complex and can be done more quickly by computer. Models can be used to investigate phenomena quickly and without ethical and practical limitations, but their usefulness is limited by how accurately the model represents the real world.	4. use a variety of models (including representational, spatial, descriptive, computational and mathematical models) to: – solve problems – make predictions – develop scientific explanations and understanding – identify limitations of models	*Examples of models:* Radiation model of light (P1.2); Wave model of light (P1.3); Physical analogies of electric circuits (P2.2, P2.3) Equations of motion(P4.2) Atomic model (P5.1) Particle model of matter (P6.1, P6.2)

IaS4 How do science and technology impact society?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Science and technology provide people with many things that they value, and which enhance their quality of life. However some applications of science can have unintended and undesirable impacts on the quality of life or the environment. Scientists can devise ways of reducing these impacts and of using natural resources in a sustainable way (at the same rate as they can be replaced). Everything we do carries a certain risk of accident or harm. New technologies and processes can introduce new risks. The size of a risk can be assessed by estimating its chance of occurring in a large sample, over a given period of time. To make a decision about a course of action, we need to take account of both the risks and benefits to the different individuals or groups involved. People are generally more willing to accept the risk associated with something they choose to do than something that is imposed, and to accept risks that have short-lived effects rather than long-lasting ones. **People's perception of the size of a particular risk may be different from the statistically estimated risk**. People tend to over-estimate the risk of unfamiliar things (like flying as compared with cycling), and of things whose effect is invisible or long-term (like ionising radiation). Some forms of scientific research, and some applications of scientific knowledge, have ethical implications. In discussions of ethical issues, a common argument is that the right decision is one which leads to the best outcome for the greatest number of people. Scientists must communicate their work to a range of audiences, including the public, other scientists, and politicians, in ways that can be understood. This enables decision-making based on information about risks, benefits, costs and ethical issues.	1. describe and explain everyday examples and technological applications of science that have made significant positive differences to people's lives 2. identify examples of risks that have arisen from a new scientific or technological advance 3. for a given situation: – identify risks and benefits to the different individuals and groups involved – discuss a course of action, taking account of who benefits and who takes the risks – suggest reasons for people's willingness to accept the risk – **distinguish between perceived and calculated risk** 4. suggest reasons why different decisions on the same issue might be appropriate in view of differences in personal, social, economic or environmental context, and be able to make decisions based on the evaluation of evidence and arguments 5. distinguish questions that could in principle be answered using a scientific approach, from those that could not; where an ethical issue is involved clearly state what the issue is and summarise the different views that may be held 6. explain why scientists should communicate their work to a range of audiences.

Linked learning opportunities

Positive applications of science:

use of the electromagnetic spectrum (P1.2);

development of electromagnetism and electric motors (P2.4, P2.5); generating and distributing electricity (P3.3);

road safety (P4.3);

Sustainability:

energy demands and choices of sources to generate electricity (P3.2)

Risks, benefits and ethical issues:

biodiversity (B6.4) technologies that use ionising radiation (P1.2, P5.2);

energy sources to generate electricity (P3.2, **P3.3, P5.3);** car safety (P4.3);

use of ionising radiation to treat disease (P5.2)